# Deflection of a simply supported beam under uniform load

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Čeština · English · Deutsch

## Introduction

This study deals with the deflection of a simply supported beam of span $L$ carrying a uniformly distributed load $q$. The aim is to compare the closed analytical solution with a numerical solution obtained by the finite-difference method[1] and to show under which conditions the Bernoulli–Navier hypothesis is sufficient. Back in the 1960s such values were read from tables; today a script of a few dozen lines does the job.

The maximum deflection at midspan is given by

$$w_{\max} = \frac{5\,q\,L^4}{384\,E\,I}, \tag{1}$$

where $E$ is Young's modulus and $I$ the second moment of area. For steel, $E = 210\,\mathrm{GPa}$; timber is roughly ten times softer. Codes usually limit the relative deflection $w_{\max}/L$ to $1/250$.

Formula (1) rests on the Bernoulli–Navier hypothesis, which neglects the shear contribution to deflection. What governs its magnitude is not slenderness as such but the **ratio of bending to shear stiffness**: for a uniform load,

$$\frac{w_{\mathrm{V}}}{w_{\mathrm{M}}} = \frac{q L^{2}/(8\,G\,A_{\mathrm{s}})}{5\,q\,L^{4}/(384\,E\,I)} = 9.6\,\frac{E\,I}{G\,A_{\mathrm{s}}\,L^{2}}, \tag{2}$$

where $G$ is the shear modulus and $A_{\mathrm{s}}$ the shear area. Only for a homogeneous rectangular section does this reduce to $\approx 2.5\,(h/L)^{2}$, so that for $h/L = 1/20$ the shear part is about 0.6 % —which is where the popular slenderness rule of thumb comes from. For sandwich, thin-walled or composite sections with a low $G\,A_{\mathrm{s}}$, shear may matter even for a very slender beam and Timoshenko theory is appropriate.

## Input data and parametric study

Table 1 summarises a parametric study of 36 combinations of span and load. The table deliberately runs over one page to exercise the repeated table header in print.

**Table 1:** Parametric deflection study—36 combinations of span and load

| no. | L [m] | q [kN/m] | E [GPa] | I [10⁻⁶ m⁴] | w_max [mm] | w/L [–] |
|---|---|---|---|---|---|---|
| 1 | 3.0 | 5.0 | 210 | 83.3 | 0.60 | 1/4970 |
| 2 | 3.0 | 10.0 | 210 | 83.3 | 1.21 | 1/2485 |
| 3 | 3.0 | 15.0 | 210 | 83.3 | 1.81 | 1/1657 |
| 4 | 3.5 | 5.0 | 210 | 83.3 | 1.12 | 1/3133 |
| 5 | 3.5 | 10.0 | 210 | 83.3 | 2.23 | 1/1567 |
| 6 | 3.5 | 15.0 | 210 | 83.3 | 3.35 | 1/1044 |
| 7 | 4.0 | 5.0 | 210 | 83.3 | 1.90 | 1/2100 |
| 8 | 4.0 | 10.0 | 210 | 83.3 | 3.81 | 1/1050 |
| 9 | 4.0 | 15.0 | 210 | 83.3 | 5.71 | 1/700 |
| 10 | 4.5 | 5.0 | 210 | 83.3 | 3.05 | 1/1475 |
| 11 | 4.5 | 10.0 | 210 | 83.3 | 6.10 | 1/738 |
| 12 | 4.5 | 15.0 | 210 | 83.3 | 9.15 | 1/492 |
| 13 | 5.0 | 5.0 | 210 | 83.3 | 4.65 | 1/1075 |
| 14 | 5.0 | 10.0 | 210 | 83.3 | 9.30 | 1/538 |

| no. | L [m] | q [kN/m] | E [GPa] | I [10⁻⁶ m⁴] | w_max [mm] | w/L [–] |
|---|---|---|---|---|---|---|
| 15 | 5.0 | 15.0 | 210 | 83.3 | 13.95 | 1/358 |
| 16 | 5.5 | 5.0 | 210 | 83.3 | 6.81 | 1/808 |
| 17 | 5.5 | 10.0 | 210 | 83.3 | 13.62 | 1/404 |
| 18 | 5.5 | 15.0 | 210 | 83.3 | 20.43 | 1/269 |
| 19 | 6.0 | 5.0 | 210 | 83.3 | 9.64 | 1/622 |
| 20 | 6.0 | 10.0 | 210 | 83.3 | 19.28 | 1/311 |
| 21 | 6.0 | 15.0 | 210 | 83.3 | 28.93 | 1/207 |
| 22 | 6.5 | 5.0 | 210 | 83.3 | 13.29 | 1/489 |
| 23 | 6.5 | 10.0 | 210 | 83.3 | 26.57 | 1/245 |
| 24 | 6.5 | 15.0 | 210 | 83.3 | 39.86 | 1/163 |
| 25 | 7.0 | 5.0 | 210 | 83.3 | 17.86 | 1/392 |
| 26 | 7.0 | 10.0 | 210 | 83.3 | 35.71 | 1/196 |
| 27 | 7.0 | 15.0 | 210 | 83.3 | 53.57 | 1/131 |
| 28 | 7.5 | 5.0 | 210 | 83.3 | 23.54 | 1/319 |
| 29 | 7.5 | 10.0 | 210 | 83.3 | 47.08 | 1/159 |
| 30 | 7.5 | 15.0 | 210 | 83.3 | 70.62 | 1/106 |
| 31 | 8.0 | 5.0 | 210 | 83.3 | 30.48 | 1/262 |
| 32 | 8.0 | 10.0 | 210 | 83.3 | 60.95 | 1/131 |
| 33 | 8.0 | 15.0 | 210 | 83.3 | 91.43 | 1/87 |
| 34 | 8.5 | 5.0 | 210 | 83.3 | 38.84 | 1/219 |
| 35 | 8.5 | 10.0 | 210 | 83.3 | 77.68 | 1/109 |
| 36 | 8.5 | 15.0 | 210 | 83.3 | 116.52 | 1/73 |

Results for spans above 8.0 m clearly violate the 1/250 limit and call for a deeper section or a camber.

## Computational script

The analytical solution is verified by the following script. The lines are deliberately long to exercise code wrapping in print.

```python
"""Simply supported beam deflection: analytical vs. numerical (finite
  differences)."""

import numpy as np


def analytical_solution(L: float, q: float, E: float, I: float, n:
  int = 101) -> np.ndarray:

    """Return the deflection curve w(x) = q x (L^3 - 2 L x^2 + x^3) /
  (24 E I)."""

    x = np.linspace(0.0, L, n)

    return q * x * (L**3 - 2.0 * L * x**2 + x**3) / (24.0 * E * I)


def numerical_solution(L: float, q: float, E: float, I: float, n: int
  = 101) -> np.ndarray:

    """Solve E I w'''' = q by finite differences with w(0) = w(L) =
  0, w''(0) = w''(L) = 0."""

    h = L / (n - 1)

    A = np.zeros((n, n)); b = np.full(n, q * h**4 / (E * I))

    for i in range(2, n - 2):

        A[i, i-2:i+3] = [1.0, -4.0, 6.0, -4.0, 1.0]

    A[0, 0] = A[n-1, n-1] = 1.0; b[0] = b[n-1] = 0.0
  # w = 0 at supports

    A[1, 0:3] = [1.0, -2.0, 1.0]; b[1] = 0.0
  # w'' = 0 (left hinge)

    A[n-2, n-3:n] = [1.0, -2.0, 1.0]; b[n-2] = 0.0
  # w'' = 0 (right hinge)

    return np.linalg.solve(A, b)
```

```
if __name__ == "__main__":

    L, q, E, I = 6.0, 10_000.0, 210e9, 83.3e-6

    w_a = analytical_solution(L, q, E, I)

    w_n = numerical_solution(L, q, E, I)

    print(f"analytical w_max = {1000 * w_a.max():.3f} mm, numerical
  w_max = {1000 * w_n.max():.3f} mm")

    print(f"relative deviation = {abs(w_n.max() - w_a.max()) /
  w_a.max():.2e}")
```

## Graphical results

The deflection curves of both solutions are plotted in Fig. 1.

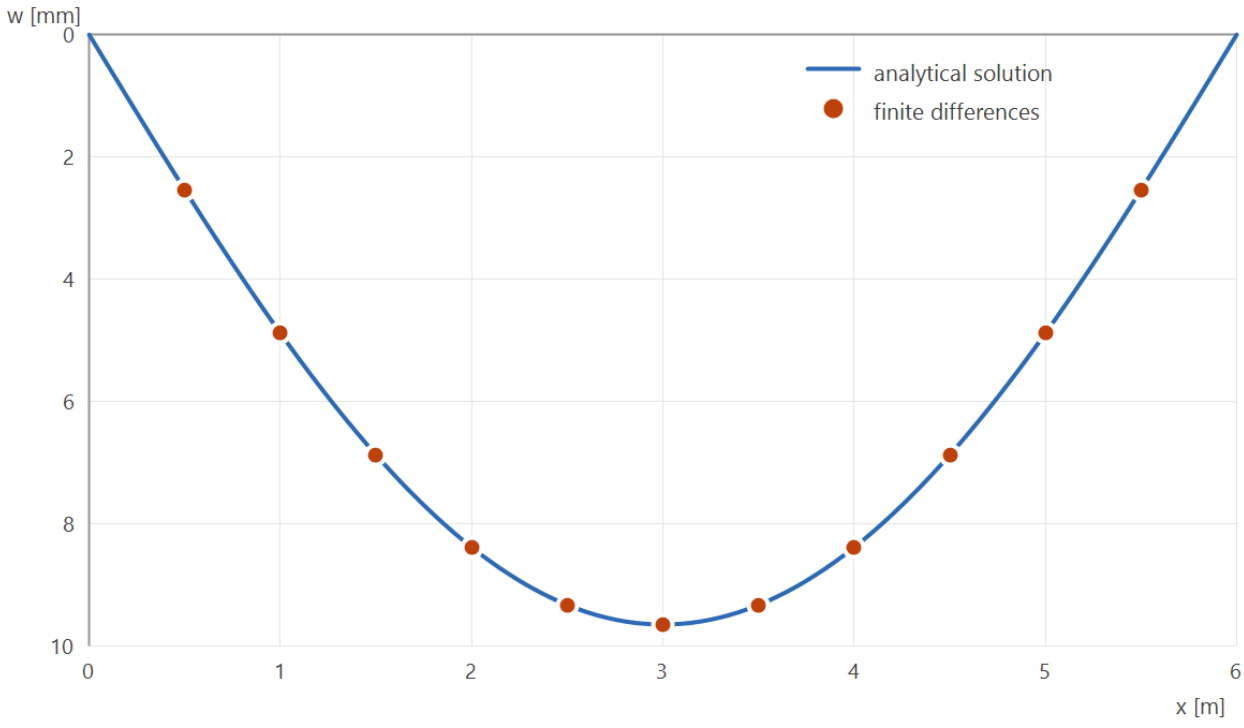



**Fig. 1:** Deflection curve—analytical vs. numerical solution

The static scheme (Fig. 2), stored under the same file name in a different folder:

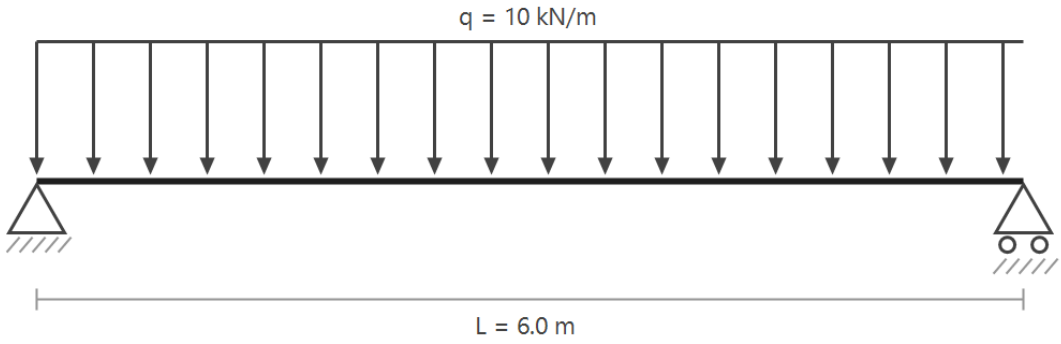



**Fig. 2:** Static scheme of the simply supported beam

## Source vs. result

This document was produced from plain Markdown. The block below is the literal source text (typographic rules are deliberately not applied inside code blocks):

```
She said "the span is 6 m" - and she's right: pages 10-20 of the
  report

list a 40x60 mm section tested at 25 °C... Don't forget section 3.
```

And this is how mdprint typesets the very same text:

She said “the span is 6 m”—and she’s right: pages 10–20 of the report list a 40×60 mm section tested at 25 °C… Don’t forget section 3.

The difference lives in the details that decide legibility: curly quotes, an em dash, an en dash in the range, a real multiplication sign, a proper apostrophe and a bound number–unit pair. Math is typeset; code and paths stay untouched.

## Conclusion

The numerical solution differs from the analytical one by less than 0.01 %, confirming both approaches and formula (1); the shear contribution per equation 2 is negligible at this slenderness. The inline relation $w(x) = \frac{qx(L^3 - 2Lx^2 + x^3)}{24EI}$ can therefore be used for quick checks[2]. A definition to close with:

**Serviceability limit state**

A state beyond which the structure no longer meets its operational requirements, although no failure occurs.

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1. Finite-difference method—the derivatives in the governing differential equation are replaced by difference quotients on a regular grid of nodes. Not to be confused with the finite element method, which seeks an approximate solution in a weak formulation over element basis functions. ↩
2. A design check per EN 1990 requires load combinations, which this text omits for brevity. ↩

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